

SQL Data Analysis Report

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This technical report documents the complete end-to-end execution of Project 3: SQL Data Analysis for the DecodeLabs Data Analytics Virtual Internship. The project required setting up a local database schema in MySQL Workbench, ingesting raw e-commerce transaction data, and executing analytical SQL queries to derive key commercial performance metrics.

The underlying dataset contains 1,200 unique sales records. Through structured SQL queries, critical insights were extracted regarding gross revenue, product popularity, marketing channel acquisition efficiency, and order fulfillment bottleneck risks.

1,200	\$1,264,761.96	3,535	\$1,053.97
Total Transactions	Gross Revenue	Total Units Sold	Average Order Value

The transactional data was imported into MySQL Server using MySQL Workbench's Table Data Import Wizard. The imported table was originally generated with the name `project 3 data analysis internship`. To align with standard relational database naming conventions and eliminate white space syntax issues, the table was formally renamed to `e_commerce_data`.

```
-- Renaming raw imported table to standard database entity name
RENAME TABLE `project 3 data analysis internship` TO e_commerce_data;
```

Query Execution, Methodology & Outputs

Task 1: Baseline Exploration & Category Identification

Objective: Inspect the dataset structure and retrieve unique product categories and customer payment methods.

```
SELECT * FROM e_commerce_data LIMIT 10;
SELECT DISTINCT Product FROM e_commerce_data;
SELECT DISTINCT PaymentMethod FROM e_commerce_data;
```

Key Findings:

- 7 Unique Products: Monitor, Phone, Tablet, Chair, Printer, Laptop, Desk.
- 5 Unique Payment Methods: Debit Card, Online, Credit Card, Gift Card, Cash.

Task 2: Row-Level Filtering & Temporal Sorting

Objective: Filter records for orders with a status of 'Delivered' and sort by Date in descending order to analyze recently completed orders.

```
SELECT OrderID, Date, CustomerID, Product, TotalPrice
FROM e_commerce_data
WHERE OrderStatus = 'Delivered'
ORDER BY Date DESC;
```

Key Findings: Successfully isolated all 231 Delivered orders chronologically. Recent high-value fulfilled orders included ORD200837 (\$3,277.75 for a Chair) and ORD200918 (\$1,841.25 for a Laptop).

Task 3: Macro Performance Metrics Aggregation

Objective: Compute high-level business performance metrics (total orders, total items sold, gross revenue, average order value, and coupon count) using SQL aggregate functions.

```
SELECT
    COUNT(OrderID) AS total_orders,
    SUM(Quantity) AS total_items_sold,
    SUM(TotalPrice) AS total_revenue,
    AVG(TotalPrice) AS average_order_value,
    COUNT(CouponCode) AS orders_with_coupons
FROM e_commerce_data;
```

Total Orders	Units Sold	Gross Revenue (\$)	Average Order Value (\$)	Coupon Usage
1,200	3,535	\$1,264,761.96	\$1,053.97	1,200

Task 4: Product Line Revenue & Volume Analysis

Objective: Group sales data by Product category to calculate sales volume, revenue contribution, and average unit price per product.

```
SELECT
    Product,
    COUNT(OrderID) AS total_orders,
    SUM(Quantity) AS total_units_sold,
    SUM(TotalPrice) AS total_revenue,
    AVG(UnitPrice) AS avg_unit_price
FROM e_commerce_data
GROUP BY Product
ORDER BY total_revenue DESC;
```

Product Category	Total Orders	Units Sold	Gross Revenue (\$)	Avg Unit Price (\$)
Chair	178	562	\$195,620.11	\$355.66
Printer	181	542	\$195,612.61	\$351.71
Laptop	173	535	\$192,126.56	\$357.71
Tablet	179	497	\$186,568.95	\$367.68
Monitor	163	480	\$175,651.41	\$358.66
Desk	170	508	\$167,459.93	\$329.61
Phone	156	411	\$151,722.39	\$375.22

Task 5: Acquisition Channel Efficiency (HAVING Clause)

Objective: Analyze performance across customer referral sources and filter out low-revenue channels generating \$200,000 or less using HAVING SUM(TotalPrice) > 200000.

```

SELECT
    ReferralSource,
    COUNT(OrderByID) AS total_orders,
    SUM(TotalPrice) AS total_revenue
FROM e_commerce_data
GROUP BY ReferralSource
HAVING SUM(TotalPrice) > 200000
ORDER BY total_revenue DESC;

```

Referral Source	Total Orders	Total Revenue (\$)
Instagram	259	\$275,285.45
Email	250	\$261,808.55
Google	241	\$250,441.48
Facebook	228	\$250,410.90
Referral	222	\$226,815.58

Task 6: Fulfillment & Order Status Breakdown

Objective: Categorize order volume and total value across fulfillment statuses to evaluate operational performance.

```

SELECT
    OrderStatus,
    COUNT(OrderByID) AS order_count,
    SUM(TotalPrice) AS total_status_value
FROM e_commerce_data
GROUP BY OrderStatus
ORDER BY order_count DESC;

```

Order Status	Order Count	Total Value (\$)	% of Volume
Cancelled	250	\$276,396.21	20.8%
Returned	247	\$243,277.70	20.6%
Pending	237	\$256,328.15	19.8%
Shipped	235	\$246,159.58	19.6%
Delivered	231	\$242,600.32	19.2%

Strategic Insights & Business Recommendations

- Address High Cancellation & Return Rates: Cancelled and Returned orders account for 497 orders worth \$519,673.91 (over 41% of potential gross revenue). Implementing stricter post-order confirmation workflows and investigating post-delivery product defects can recover substantial revenue.
- Prioritize Instagram & Email Marketing Channels: Instagram (\$275.2k) and Email (\$261.8k) lead all customer acquisition channels. Reallocating digital ad budgets toward Instagram visual campaigns will deliver the highest return on ad spend (ROAS).

3. Leverage Top Product Revenue Categories: Chairs (\$195.6k), Printers (\$195.6k), and Laptops (\$192.1k) generated the highest revenue. Creating promotional bundles pairing Chairs with Desk setups or Laptops with Printers will further boost Average Order Value (AOV).